

CarlQ's Guide To DATA-DRIVEN INTELLIGENT CARS



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In God we trust, all others bring data.” When there is so much data around, the challenge is to make information from such resources or objects more meaningful. Now with the Internet of Things (IoT), an ocean of data can be captured once connectivity barriers are surpassed, which is why CarlQ has proved that even cars can bring a wealth of information.

Why the car

Data received from connected cars can prove a huge potential from improving mileage to receiving suggestions on where your next road trip could be. Developers in the IoT zone are facing challenges to meet such demand for high-performance IoT

Inside CarlQ

- Chip to decode ECU communication
- GSM for transmitting data to the Cloud
- GPS for location tracking
- Onboard storage in case the car is out of coverage area
- Accelerometer for measuring impact

applications while keeping these simple and effective. With possibilities of even your toothbrush getting connected to the Internet becoming inevitable, cars cannot be out of the crosshairs. A car has so much data that it can tell us about itself and everything that controls it, which makes it quite interesting.

“We first explored if it was possible to read data from a car, and once we figured out that it was, CarlQ began its journey, serving its purpose of making this data available to anyone and bettering the life of a car owner,” says Deepak Thomas, co-founder at CarlQ.

CarlQ uses an IoT device called Smart-Plug to read data from the car. It fits snugly into the onboard diagnostics port of the car and works as a simple plug-n-play device. Most people are unaware that onboard diagnostics is a standard port on all cars and is located under the steering wheel. Once connected, the car's data is sent to Cloud servers, where it is analysed with hundreds of similar cars, and consumable data is given back to the users. They can set custom alerts for speeding/geo-fence, share trips on social media, view vital stats about the cars and connect with car-ecosystem players like workshops and roadside-assistance providers.

Transformational integration of the platform

Looking at the insides of CarlQ, we find that real-time communication and proprietary components need to be tested, so as to ensure seamless connectivity when



Fig. 1: CarlQ device



Fig. 2: CarlQ dashboard